Free energy of the population chart integral: $-\log Z(n) = \frac{p}{2}\log n - m\log\log n + F_0 + o(1)$

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Abstract

Taking logarithms in the multiplicity theorem of unit 59 gives the free-energy form familiar from singular learning theory. Generically, $Z \sim C\,n^{-\lambda}(\log n)^m$ with $C > 0$ implies $\log Z + \lambda \log n - m \log\log n \to \log C$; applied to the box integral this yields $-\log \int_{(0,b]^{D+2}} u^h f(\sqrt{n}\,u^k)\,du = \frac{p}{2}\log n - m\log\log n + F_0 + o(1)$ with $p/2$ half the minimal candidate exponent and $m+1$ its multiplicity, for any exponent vector. Zero `sorry`/`axiom`.

1 Setting

An asymptotic equivalence $Z \sim v$ with v eventually nonzero is the statement $Z/v \to 1$; since log is continuous at 1, $\log(Z/v) \to 0$, and $\log v$ is computed exactly for $v = Cn^{-\lambda}(\log n)^m$ when $n > 1$.

2 The things formalised

```
theorem tendsto_log_of_isEquivalent_powLog {Z :   →  } {C lam :  } {m :  } (hC : 0 < C)
    (hZ : Z ~[atTop] fun n => C * (n ^ (-lam) * Real.log n ^ m)) :
    Tendsto (fun n => Real.log (Z n) + lam * Real.log n - m * Real.log (Real.log n))
      atTop (  (Real.log C))
theorem boxIntegralFin_freeEnergy (  a b :  ) (h  : 0 <  ) (hb : 0 < b) (D :  )
    (k h : Fin (D + 2) →  ) (hk :   i, 0 < k i) :
      (p :  ) (m :  ) (F  :  ), (  i, p   finExp k h i)
      m + 1 = (Finset.univ.filter fun i => finExp k h i = p).card
      Tendsto (fun n => -Real.log (boxIntegralFin   a b (Real.sqrt n) k h)
        - (p / 2 * Real.log n - m * Real.log (Real.log n))) atTop (  F )
```

3 Proof strategy

`isEquivalent_iff_tendsto_one` gives $Z/v \to 1$; `Filter.Tendsto.log` gives $\log(Z/v) \to 0$; eventually $Z \neq 0$ (as $Z/v > 0$), so $\log(Z/v) = \log Z - \log v$ with $\log v = \log C - \lambda \log n + m \log\log n$ by `Real.log_mul`, `Real.log_rpow`, `Real.log_pow`. The free-energy statement negates and rearranges.

4 Roadblocks and resolutions

- None: first-pass compile.

5 What was Mathlib, what was new

Mathlib: `isEquivalent_iff_tendsto_one`, `Filter.Tendsto.log`, `lt_mem_nhds`, `Real.log_div`, `Real.log_rpow`. New: the generic logarithm lemma and the chart free energy.

6 Lessons

The free-energy form is one lemma away from the asymptotic equivalence; stating it makes the connection between the chart integrals and the SLT quantities $\lambda \log n - (m-1) \log \log n$ explicit.

7 Outcome

The population chart free energy has the Watanabe form with the exponent and multiplicity read off from (k, h) by the candidate-exponent rule, in every dimension. This closes the constant-ξ programme (units 44–61) on the free-energy side as well.